

Minjae Kim

Senior Researcher, Electronics and Telecommunications Research Institute (ETRI)

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Research theme: Advancing human ability through robotics and assistive systems.

Robotic control, medical and assistive robotics, prosthetic control, haptics, and wearable sensing systems.

Profile

Robotics researcher and system designer whose work centers on advancing human ability through medical robots, assistive devices, prosthetic control, wearable sensing, and real-world robotic systems. My work combines control architecture, embedded systems, sensor integration, machine learning, and clinical or user-facing validation.

Research Interests

- Robotic control and system design
- Neural and myoelectric interfaces
- Haptics and wearable sensing systems
- Assistive and medical robotic systems
- AI-based prosthetic and rehabilitation control

Experience

Electronics and Telecommunications Research Institute (ETRI)

Senior Researcher

Jul 2026 - Present

- Research on robotic systems and assistive technologies.

Neuromeka

Leader / Group Leader, Medical Robot Group

Jan 2026 - Jun 2026

Manager / Group Leader, HRI Group

Apr 2024 - Jan 2026

Robotic Systems Integration and Control

- Designed and implemented control architectures for industrial and service robotic applications, including a frying chicken robotic platform, four-wheeled mobile robot, round pallet system, and Cobosol platform.
- Led system integration for a space robotic arm, surgical robotic systems, and humanoid robotic platforms for surgical assistance.
- Directed cross-functional development toward GMP compliance, medical device certification, safety validation, and full-system testing.
- Built digital-twin and integrated control environments using NVIDIA Isaac Sim, robotic manipulators, and multi-finger robot hands.

March Lab, Pohang University of Science and Technology

Post-Doctoral Fellow

Supervisor: Dr. Keehoon Kim

Jun 2023 - Apr 2024

Wearable Bio-signal Systems

- Designed and developed a compact EMG sensor module under 20 mm x 20 mm with integrated circuitry, PCB, microcontroller, and communication capabilities.
- Implemented BLE-based real-time communication for bio-signal acquisition and analysis.
- Developed Android and iOS applications for signal monitoring and analysis.

Center for Bionic Medicine, Shirley Ryan AbilityLab

Post-Doctoral Fellow

Jan 2021 - May 2023

Visiting Post-Doctoral Scholar, supported by KIST and Hanyang University

Jan 2020 - Dec 2020

Supervisor: Dr. Levi Hargrove

Prosthetics and AI-based Control

- Developed deep neural network models to estimate impedance control parameters from prosthetic sensor data, including IMU and load cell measurements.
- Established a real-time prosthetic leg control framework and conducted experiments with disabled patients.
- Built hardware and software interfaces to deploy control models on an Android-based embedded system for real-time prosthetic operation.
- Trained models on gait data from able-bodied subjects for intent recognition and prosthetic control in disabled patients.
- Generated synthetic gait trajectories using generative adversarial networks to create unseen motions beyond the training dataset.
- Designed selective artifact removal methods for sEMG signals recorded during spinal stimulation experiments with stroke patients.

Center for Intelligent and Interactive Robotics, KIST

Post-Doctoral Fellow

Supervisor: Dr. Keehoon Kim

Sep 2018 - Aug 2019

- Developed a microneedle array-based sEMG system by integrating embedded systems and electrodes.
- Developed sEMG signal-based motion recognition methods.
- Proposed model-based compensation for electrode displacement and robust motion recognition independent of electrode properties.
- Generated synthetic sEMG signals using model-based approaches.

Robotics Lab, Pohang University of Science and Technology

Ph.D. Student

Supervisor: Dr. Wan Kyun Chung

Mar 2012 - Aug 2018

- Designed and fabricated polymer- and metal-based microneedle array electrodes, including high-density configurations.
- Evaluated electrode performance under multiple experimental conditions and analyzed parameters affecting signal quality.
- Designed a complete wireless wearable sEMG system pipeline, including electrode design, circuit design, and casing.
- Proposed feature extraction methods for robust motion recognition and validated the model with healthy subjects and an amputee participant.
- Developed a cell stimulation and signal measurement testbed using an XYZ micro-stage for precise cell manipulation.
- Integrated sensors including load cells and thermistors for prosthetic hand evaluation.

Intelligent Mechatronics and Robotics Lab, Chung-Ang University

Summer Intern

Advisor: Dr. Tae-Hyoung Kim

Jul 2011 - Aug 2011

- Studied Constrained Particle Swarm Optimization through literature review.
- Implemented the CPSO algorithm from scratch in MATLAB.

Education

Pohang University of Science and Technology

Combined M.S./Ph.D. in Mechanical Engineering

Pohang, Korea

Mar 2012 - Aug 2018

- Supervisor: Dr. Wan Kyun Chung
- Dissertation: *Development of sEMG Interface and Algorithm for Long-term Reliable Measurement*

Chung-Ang University

B.S. in Mechanical Engineering

Seoul, Korea

Mar 2008 - Feb 2012

- Minor in Electrical and Electronics Engineering

Honors and Awards

- Sarah Baskin Outstanding Research Award, 1st place, Shirley Ryan AbilityLab, Jun 2022

- RMI Paper Award, Korea Institute of Science and Technology, Dec 2019
- Outstanding Paper Award, International Conference on Ubiquitous Robots, Jun 2018
- Best Paper Award, Korean Conference on Sensors, Nov 2015
- Best Teaser Award, Haptic Symposium, Feb 2014
- Participation Award, Information Search Competition, Chung-Ang University, Nov 2011

Journal Publications

1. **Minjae Kim** and L. J. Hargrove, "A Method for Controlling Prosthetic Legs Using Deep Neural Network Models Trained On Synthetic Gait Trajectory," *Journal of NeuroEngineering and Rehabilitation*, Vol. 20:115, 1-16, Sep 2023.
2. S. Yang, **Minjae Kim**, S. K. Hong, S. Kim, W. K. Chung, G. Lim, and H. Jeon, "Design of 3D Controller Using Nanocracking Structure-Based Stretchable Strain Sensor," *Sensors*, Vol. 23, 1-14, May 2023.
3. **Minjae Kim** and L. J. Hargrove, "A Gait Phase Prediction Model Trained on Benchmark Datasets For Evaluating a Controller For Prosthetic Legs," *Frontiers in Neurorobotics*, Vol. 16, 1-13, Jan 2023.
4. **Minjae Kim**, A. M. Simon, and L. J. Hargrove, "Seamless and intuitive control of a powered prosthetic leg using deep neural network for transfemoral amputees," *Wearable Technologies*, Vol. 3, e24:1-16, Sep 2022.
5. **Minjae Kim** and L. J. Hargrove, "Deep-Learning to Map a Benchmark Dataset of Non-amputee Ambulation for Controlling an Open Source Bionic Leg," *IEEE Robotics and Automation Letters*, Vol. 7, Issue 4, 10597-10604, Jul 2022.
6. **Minjae Kim**, Y. Moon, J. Hunt, K. A. McKenzie, A. Horin, M. McGuire, K. Kim, L. J. Hargrove, and A. Jayaraman, "A Novel Technique to Reject Artifact Components for Surface EMG Signals Recorded During Walking With Transcutaneous Spinal Cord Stimulation: A Pilot Study," *Frontiers in Human Neuroscience*, Vol. 15, Jun 2021.
7. **Minjae Kim**, W. K. Chung, and K. Kim, "Subject-Independent sEMG Pattern Recognition by Using a Muscle Source Activation Model," *IEEE Robotics and Automation Letters*, Vol. 5, Issue 4, 5175-5180, Jul 2020.
8. **Minjae Kim**, K. Kim, and W. K. Chung, "Simple and Fast Compensation of sEMG Interface Rotation for Robust Hand Motion Recognition," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, Vol. 26, Issue 12, 2397-2406, Dec 2018. Selected as the front cover of the 12th issue in the 26th volume.
9. **Minjae Kim** and W. K. Chung, "Spatial sEMG Pattern-Based Finger Motion Estimation in a Small Area Using a Microneedle-Based High-Density Interface," *IEEE Robotics and Automation Letters*, Vol. 3, No. 1, 234-241, Jan 2018.
10. **Minjae Kim**, G. Gu, K. J. Cha, D. S. Kim, and W. K. Chung, "Wireless sEMG system with a microneedle-based high-density electrode array on a flexible substrate," *Sensors*, Vol. 18, No. 1, 92, Dec 2017.
11. **Minjae Kim**, T. Kim, D. S. Kim, and W. K. Chung, "Curved microneedle array-based sEMG electrode for robust long-term measurements and high selectivity," *Sensors*, Vol. 15, No. 7, 16265-16280, Jul 2015.

Conference Publications

1. D. G. Park, Y. J. Lee, J. B. Han, **Minjae Kim**, and T. K. Yeu, "A Preliminary Study on Fatigue Estimation for Underwater Robot Operator Using sEMG Signals," *IEEE International Conference on Automation Science and Engineering (CASE)*, New Zealand, Aug 2023.
2. **Minjae Kim**, A. M. Simon, K. Shah, and L. J. Hargrove, "Machine Learning-Based Gait Mode Prediction for Hybrid Knee Prosthesis Control," *45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1-6, Australia, Jul 2023.
3. **Minjae Kim**, W. K. Chung, and K. Kim, "Motion Intensity Extraction Scheme for Simultaneous Recognition of Wrist/Hand Motions," *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10112-10117, Virtual, May 2020.
4. **Minjae Kim**, W. K. Chung, and K. Kim, "Preliminary Study of Virtual sEMG Signal-Assisted Classification," *International Conference on Rehabilitation Robotics (ICORR)*, pp. 1133-1138, Canada, Jun 2019.
5. **Minjae Kim** and W. K. Chung, "Muscle Activation Source Model-based sEMG Signal Decomposition and Recognition of Interface Rotation," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 2780-2786, Spain, Oct 2018.
6. **Minjae Kim**, G. Gu, and W. K. Chung, "Pneumatic Microneedle-Based High-Density sEMG Sleeve for Stable and Comfortable Skin Contact during Dynamic Motion," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 5477-5482, Spain, Oct 2018.
7. **Minjae Kim**, G. Gu, and W. K. Chung, "Pneumatic Sleeve-Assisted Stable sEMG Measurement for Microneedle Array Electrode," *International Conference on Ubiquitous Robots (UR)*, pp. 1-4, USA, Jun 2018.
8. **Minjae Kim**, G. Gu, and W. K. Chung, "Skin Grasping sEMG Interface Based on Microneedle Array Electrode," *International Conference on Ubiquitous Robots and Ambient Intelligence (URAI)*, pp. 839-840, Korea, Jun 2017.
9. **Minjae Kim**, D. S. Kim, and W. K. Chung, "Microneedle-based high-density surface EMG interface with high selectivity for finger movement recognition," *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 973-978, Sweden, May 2016.
10. J. Lee, E. Cho, **Minjae Kim**, Y. Yoon, and S. Choi, "PreventFHP: Detection and Warning System for Forward Head Posture," *Haptic Symposium*, pp. 295-298, USA, Feb 2014.
11. **Minjae Kim**, S. Park, and W. K. Chung, "Flexible Polymer-based Micro Needle Array sEMG Sensor," *International Conference on Ubiquitous Robots and Ambient Intelligence (URAI)*, pp. 1-4, Korea, Oct 2013.

Non-refereed Publications

1. G. Jeong, Y. Kim, Y. Oh, G. Gu, S. Park, **Minjae Kim**, J. An, and K. Kim, "Design and analysis of underactuated prosthetic hand using slider-crank mechanism," Korea Robotics Society Annual Conference, Jan 2019.
2. G. Gu, **Minjae Kim**, and W. K. Chung, "Development of Wireless High-Density sEMG Acquisition System," Korea Robotics Society Annual Conference, Jan 2018.
3. **Minjae Kim** and W. K. Chung, "sEMG Source Activation Model-Based Motion Estimation Robust to Electrode Placement Change," Korea Robotics Society Annual Conference, Jan 2018.
4. **Minjae Kim** and W. K. Chung, "Novel sEMG-Muscle Model-based Approach for Robust Motion Identification," The Sixth Asia International Symposium on Mechatronics, Sep 2017.
5. G. Gu, **Minjae Kim**, and W. K. Chung, "Design of Portable sEMG Sensor System," Korea Robotics Society Annual Conference, Feb 2017.
6. **Minjae Kim**, T. Kim, D. S. Kim, and W. K. Chung, "Metal Microneedle-based Multi-channel Interface for Biosignal Measurement," Korean Conference on Sensors, Nov 2015.
7. **Minjae Kim**, D. S. Kim, and W. K. Chung, "Reliability of surface electromyographic measurement using microneedle array electrode," International Conference on Biofabrication in Korea, Sep 2014.
8. **Minjae Kim**, T. Kim, D. S. Kim, and W. K. Chung, "Biocompatible microneedle array electrode for surface electromyography," Korean Society of Mechanical Engineers conference of bioengineering, Apr 2014.
9. **Minjae Kim** and W. K. Chung, "ICRA 2013 Review," Journal of Korea Robotics Society, Oct 2013.
10. **Minjae Kim** and W. K. Chung, "The comparison of the sEMG pattern classification performance depends on the electrode impedance," Korea Robotics Society Annual Conference, May 2013.